

Probability Theory

Labix

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Abstract

Notes for the basics of Probability Theory.

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1 Foundations of Probability Theory

1.1 Definition of Probability

Definition 1.1.1 (Probability Space)

A probability space is a measure space $(\Omega, \mathcal{F}, P : \mathcal{F} \rightarrow [0, 1])$.

Explicitly, a probability space is a triple $(\Omega, \mathcal{F}, P)$ consisting of the following data:

- $\Omega \neq \emptyset$ is a set called the sample space.
- $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra called events.
- $P : \mathcal{F} \rightarrow [0, 1]$ is a set function.

such that the following are true:

- $P(\Omega) = 1$.
- If $\{A_n \mid n \in \mathbb{N}\} \subseteq \mathcal{F}$ are pairwise disjoint, then

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k)$$

Definition 1.1.2 (Events and Probability)

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

- An event of the space is an element of $\mathcal{F}$.
- Let $A \in \mathcal{F}$ be an event. The probability of A is defined to be the value $P(A)$.

Example 1.1.3

Let Ω be the set of all possible outcomes of flipping a coin ten times. Let $\mathcal{F} = \mathcal{P}(\Omega)$ be the set of subsets of Ω . Let $P : \mathcal{F} \rightarrow [0, 1]$ be the function defined by $P(E) = \frac{|E|}{|\Omega|}$ for all $E \in \mathcal{F}$. Then $(\Omega, \mathcal{F}, P)$ is a probability space.

Proposition 1.1.4

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. Then the following are true.

- $P(\Omega \setminus A) = 1 - P(A)$
- $A \subset B \implies P(A) \leq P(B)$

Proof

Let $A \subset B \subset \Omega$ be events in Ω .

- A and $\Omega \setminus A$ are disjoint and $P(\Omega) = P(A) + P(\Omega \setminus A)$ and $P(\Omega \setminus A) = 1 - P(A)$
- We have that A and $B \setminus A$ are disjoint. Thus $P(B) = P(A) + P(B \setminus A)$. Since $P(B \setminus A) \geq 0$, we have $P(A) \leq P(B)$.

■

Example 1.1.5 (The Birthday Problem)

Suppose that there are k people in a room, all of whom are equally likely to have a birthday in one of the 365 days in a year. Suppose that the birthdays of the people in the room are independent of each other. What is the probability that two or more people in the room have the same birthday?

Proof

Let $\Omega = \{\{n_1, \dots, n_k\} \mid 1 \leq n_1, \dots, n_k \leq 365\}$. These indicate all the possible birthday configurations. Let $\mathcal{F} = \mathcal{P}(\Omega)$. This means that every birthday configuration we constructed is possible.

By the assumptions of the question, we set $P(E) = \frac{|E|}{|\Omega|}$ for an event $E \in \mathcal{F}$. Then we have

$$P(\text{at least one birthday match}) = 1 - P(\text{no birthday matches}) = 1 - \frac{365!}{(365)^k(365 - k)!}$$

■

Theorem 1.1.6 (Principle of Inclusion Exclusion)

Let $A, B \subset \Omega$ be a sample space and P the probability measure.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof

Note that

$$\begin{aligned} A \cup (B \setminus A) &= A \cup (B \cap A^c) \\ &= (A \cup B) \cap (A \cup A^c) \\ &= A \cup B \end{aligned}$$

Note also that $A \cap (B \setminus A) = \emptyset$. Thus $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B)$ ■

Theorem 1.1.7 (Extended Principle of Inclusion Exclusion) Let $A_k \subset \Omega$ be a sample space and P the probability measure for all $k \leq n \in \mathbb{N}$. Then

$$P\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 \leq \dots \leq i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k})$$

1.2 Conditional Probability

Definition 1.2.1 (Conditional Probability)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. Define the conditional probability of A given B to be

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Proposition 1.2.2 (Multiplication Rule)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A_1, \dots, A_n \in \mathcal{F}$ be events. Then we have

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) \cdot \prod_{i=2}^n P\left(A_i \mid \bigcap_{j=1}^{i-1} A_j\right)$$

Proof

Follow immediately by induction on the definition of conditional probability. ■

Theorem 1.2.3 (Bayes' Rule)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be spaces such that $P(A), P(B) > 0$. Then we have

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Proof

We have that $P(A \cap B) = P(A|B)P(B)$ and $P(A \cap B) = P(B|A)P(A)$. ■

Theorem 1.2.4 (Law of Total Probability)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A_1, \dots, A_n \in \mathcal{F}$ be events that partition Ω such that $P(A_i) > 0$ for $1 \leq i \leq n$. Let $B \in \mathcal{F}$ be an event. Then we have

$$P(B) = \sum_{k=1}^n P(A_k)P(B|A_k)$$

Proof

Note that since $A_1, \dots, A_n$ is a partition of Ω , $B \cap A_1, \dots, B \cap A_n$ is a partition of B . Then we have

$$\begin{aligned} \sum_{k=1}^n P(A_k)P(B|A_k) &= \sum_{k=1}^n P(B \cap A_k) \\ &= P\left(\bigcup_{k=1}^n B \cap A_k\right) \\ &= P(B \cap \Omega) \\ &= P(B) \end{aligned}$$

Example 1.2.5 (Gambler's Ruin)

A gambler starts with an initial fortune of $\pounds k$ for $1 \leq k \leq n$. On each successive gamble, the gambler either wins $\pounds 1$ or loses $\pounds 1$ independent of past games, with a probability of p and $1 - p$ respectively for $p \in (0, 1)$. What is the probability that the gambler reaches $\pounds n$ dollars before reaching $\pounds 0$?

Proof

The gambler is said to have lost the game if the gambler reaches $\pounds 0$ after any round. Let a_i denote the probability that the gambler reaches $\pounds n$ dollars before reaching $\pounds 0$, when the gambler has $\pounds i$. By the law of total probability, we obtain a recurrence relation

$$a_i = pa_{i+1} + (1 - p)a_{i-1}$$

This makes for a difference equation with boundary conditions $a_0 = 0$ and $a_n = 1$. Solving it gives

$$a_i = \begin{cases} \frac{i}{n} & \text{if } p = \frac{1}{2} \\ \frac{1 - ((1-p)/p)^i}{1 - ((1-p)/p)^n} & \text{otherwise} \end{cases}$$

Notice that the $p = 1/2$ case is consistent with the $p \neq 1/2$ cases since $\lim_{p \rightarrow 1/2} \frac{1 - ((1-p)/p)^i}{1 - ((1-p)/p)^n} = \frac{i}{n}$.

1.3 Independent Events

Definition 1.3.1 (Independent Events)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. We say that A and B are independent if

$$P(A \cap B) = P(A)P(B)$$

Proposition 1.3.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. If A and B are independent, then the following are true.

- $\Omega \setminus A$ and B are independent.
- A and $\Omega \setminus B$ are independent.
- $\Omega \setminus A$ and $\Omega \setminus B$ are independent.

2 Probability Distributions

2.1 Random Variables and its Distribution

Definition 2.1.1 (Random Variable)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. An $(E, \mathcal{E})$ valued random variable is an $\mathcal{F}$ -measurable function $X : \Omega \rightarrow E$.

Definition 2.1.2 (Independent Random Variables)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. Let $X, Y : \Omega \rightarrow E$ be random variables. We say that X and Y are independent if for any $A, B \in \mathcal{E}$, we have that $X^{-1}(A)$ and $Y^{-1}(B)$ are independent events in $\mathcal{F}$.

Definition 2.1.3 (Discrete and Continuous Random Variables)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

- We say that X is discrete if $\text{im}(X)$ is a countable subset of $\mathbb{R}$.
- We say that X is continuous otherwise.

Recall that X is an $\mathcal{F}$ -measurable function if $X^{-1}(B) \in \mathcal{F}$ for $B \in \mathcal{E}$.

Definition 2.1.4 (Probability Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. Let $X : \Omega \rightarrow E$ be a measurable function. Define the probability distribution of X to be the pushforward measure $P \circ X^{-1} = P_X : \mathcal{E} \rightarrow [0, 1]$ defined by

$$P_X(A) = P(X^{-1}(A))$$

for $A \in \mathcal{E}$.

Definition 2.1.5 (Probability Density Functions and Probability Mass Function)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

- Suppose that X is a discrete random variable. Define the probability mass function of X to be the Radon–Nikodym derivative

$$f_X = \frac{dX_*P}{d\#}$$

where $\#$ is the counting measure on $\mathbb{R}$.

- Suppose that X is a continuous random variable. Define the probability density function of X to be the Radon–Nikodym derivative

$$f_X = \frac{dX_*P}{d\mu}$$

where μ is the Lebesgue measure.

Recall that this means that f_X satisfies the property that

$$P_X(A) = \int_A f_X d\mu$$

for any measurable set $A \subseteq \mathbb{R}$. In particular, if $A = \{a\} \subseteq \mathcal{F}$, then we have

$$P_X(a) = f_X(a)$$

The probability distribution function has its input as every measurable subset of $\mathbb{R}$, while the probability density function takes input as individual points of $\mathbb{R}$. They are really the same thing because having its probability be determined on singletons is sufficient to determine the probability of every measurable subset.

Proposition 2.1.6

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then the following are true.

- Suppose that X is a discrete random variable. Then the probability density function of $Y = g \circ X$ is given by

$$f_Y(y) = \sum_{x \in g^{-1}(y)} f_X(x)$$

- Suppose that X is a continuous random variable. Suppose that g is strictly monotone and differentiable. Then the probability density function of $Y = g \circ X$ is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy}(g^{-1}(y)) \right|$$

for all $y \in \mathbb{R}$

2.2 Cumulative Density Functions

Definition 2.2.1 (Cumulative Distribution Function)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Define the cumulative distribution function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ of X to be

$$F_X(x) = P_X(X \leq x)$$

Proposition 2.2.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then the following are true.

- $f_X = \frac{dF_X}{dx}$.
- $F_X(x) = \int_{-\infty}^x f_X(t) dt$

Proposition 2.2.3

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then the following are true regarding the cumulative distribution function F_X .

- F_X is monotonically increasing: $x \leq y \implies F_X(x) \leq F_X(y)$
- F_X is right continuous: If (x_n) is a sequence such that $x_1 \geq \dots \geq x_n \geq x_{n+1} \geq \dots \geq x$ and $(x_n) \rightarrow x$, then $F_X(x_n) \rightarrow F_X(x)$
- $F_X(-\infty) = 0$ and $F_X(\infty) = 1$

Proposition 2.2.4

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then we have

$$P(\{\omega \in \Omega \mid a < X(\omega) \leq b\}) = F_X(b) - F_X(a)$$

2.3 Examples of Discrete Random Variables

Definition 2.3.1 (Discrete Uniform Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a discrete uniform distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{n} & \text{if } x = 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

for some $n \in \mathbb{N}$.

Definition 2.3.2 (Bernoulli Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a Bernoulli distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$. In this case we write $X \sim \text{Bern}(p)$.

Definition 2.3.3 (Binomial Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a binomial distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x = 0, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$ and $n \in \mathbb{N}$. In this case we write $X \sim \text{Binom}(n, p)$.

Definition 2.3.4 (Poisson Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a Poisson distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

for some $\lambda > 0$. In this case we write $X \sim \text{Poisson}(\lambda)$.

Definition 2.3.5 (Geometric Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a geometric distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} p(1-p)^x & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$. In this case we write $X \sim \text{Geom}(p)$.

2.4 Examples of Continuous Random Variables

Let $I \subseteq \mathbb{R}$ be an interval. Recall that $\mathcal{B}(I)$ refers to the borel measurable subsets of I . Denote λ the Lebesgue measure on $\mathbb{R}^n$.

Definition 2.4.1 (Continuous Uniform Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow [a, b]$ be a random variable for some $a, b \in \mathbb{R}$. We say that X has a continuous uniform distribution if the probability density function of X is given by

$$f_X(x) = \frac{1}{b-a}$$

for all $x \in [a, b]$.

Definition 2.4.2 (Normal Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has a

normal distribution if the probability density function of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for some $\mu, \sigma \in \mathbb{R}$ such that $\sigma > 0$. In this case we write $X \sim \text{Normal}(\mu, \text{sigma})$.

Definition 2.4.3 (Exponential Distribution)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. We say that X has an exponential distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

for some $\lambda > 0$. In this case we write $X \sim \text{Exp}(\lambda)$.

Definition 2.4.4 (Gamma Distribution)

2.5 Multiple Random Variables

Definition 2.5.1 (Joint Probability Density and Mass Functions)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables.

- Suppose that X and Y are discrete random variables. Define the joint probability mass function of X and Y to be the Radon-Nikodym derivative

$$f_X = \frac{(X, Y)_* P}{d\#}$$

where $\#$ is the counting measure on $\mathbb{R}^2$.

- Suppose that X and Y are continuous random variables. Define the joint probability density function of X and Y to be the Radon-Nikodym derivative

$$f_X = \frac{(X, Y)_* P}{d\mu^2}$$

where μ^2 is the Lebesgue measure on $\mathbb{R}^2$.

As with the case of pmfs with one random variable, we have

$$f_X(x, y) = P(X = x, Y = y)$$

for all $x, y \in \mathbb{R}$.

2.6 Algebra of Random Variables

Proposition 2.6.1

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Then we have

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z-w) dw$$

2.7 Conditional Probability Density and Mass Functions

Definition 2.7.1 (Conditional Probability Density and Mass Functions)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Suppose that $f_Y(y) > 0$ for some $y \in \Omega$.

- Suppose that X, Y are discrete. Define the conditional probability mass function of X given

$Y = y$ to be the function

$$f_{X|Y}(x, y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

- Suppose that X, Y are continuous. Define the conditional probability density function of X given $Y = y$ to be the function

$$f_{X|Y}(x, y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

3 Expectation and Variance

3.1 Expectations

Definition 3.1.1 (Expectations)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Define the expectation of X to be

$$E[X] = \int_{\Omega} X dP$$

Lemma 3.1.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then we have

$$E[X] = \int_{\mathbb{R}} x f_X(x) dx$$

Example 3.1.3

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable such that $X \sim \text{Binom}(n, p)$ for some $n \in \mathbb{N}$ and $p \in [0, 1]$. Then we have

$$E[X] = np$$

Proof

We have

$$\begin{aligned} E[X] &= \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=1}^n \frac{n! \cdot k}{k!(n-k)!} p^k (1-p)^{n-k} \\ &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1-p)^{n-k} \\ &= \sum_{i=0}^{n-1} n \binom{n-1}{i} p^{i+1} (1-p)^{n-i-1} & (i = k-1) \\ &= np \sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{n-i-1} \\ &= np \end{aligned}$$

■

Example 3.1.4

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable such that $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$. Then we have

$$E[X] = \frac{1}{\lambda}$$

Proof

We have

$$\begin{aligned}
 E[X] &= \int_0^\infty \frac{1}{\lambda} x e^{-x/\lambda} dx \\
 &= [-x e^{-x/\lambda}]_0^\infty + \int_0^\infty e^{-x/\lambda} dx \\
 &= \int_0^\infty e^{-x/\lambda} dx \\
 &= \lambda
 \end{aligned}$$

■

Proposition 3.1.5 (Law of the Unconscious Staticians) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be random variables. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then we have

$$E[g \circ (X_1, \dots, X_n)] = \int_{\mathbb{R}^n} g(x_1, \dots, x_n) f_{(X_1, \dots, X_n)}(x_1, \dots, x_n) dx_1 \cdots dx_n$$

Proposition 3.1.6 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Then the following are true.

- If X, Y are random variables and $a, b \in \mathbb{R}$, then

$$E[aX + bY] = aE[X] + bE[Y]$$

- If $P(X \geq Y) = 1$, then

$$E[X] \geq E[Y]$$

Proposition 3.1.7 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Then X, Y are independent if and only if

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

for any two functions $g, h : \mathbb{R} \rightarrow \mathbb{R}$.

3.2 Variance and Covariance

Definition 3.2.1 (Variance)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Define the variance of X to be

$$\text{Var}(X) = E[(X - E[X])^2]$$

Example 3.2.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable such that $X \sim \text{Exp}(\lambda)$. Then we have

$$\text{Var}(X) = \lambda^2$$

Lemma 3.2.3

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then the following are true.

- $\text{Var}(X) \geq 0$.
- $\text{Var}(X) = 0$ if and only if $P_X(E[X]) = 1$.
- $\text{Var}(X) = E[X^2] - E[X]^2$
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$ for any $a, b \in \mathbb{R}$.

Proposition 3.2.4 Suppose that $X_1, \dots, X_n$ are independent variables with finite variance. Then

$$\text{Var}\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n \text{Var}(X_k)$$

Definition 3.2.5 (Covariance) Let X, Y be two random variables. The covariance of X, Y is defined as

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Proposition 3.2.6 Suppose that X, Y are random variables.

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$
- If X, Y are independent, $\text{Cov}(X, Y) = 0$
- $\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$

Proposition 3.2.7 (Variance of Sums) For random variables $X_1, \dots, X_n$, we have

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j)$$

Theorem 3.2.8 Given two random variables X and Y , we have

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}$$

Theorem 3.2.9 (Correlation Coefficient) The correlation coefficient between two random variables X and Y is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Proposition 3.2.10 Let X and Y be random variables. We have

$$-1 \leq \rho(X, Y) \leq 1$$

Moreover, for any $a, b, c, d \in \mathbb{R}$ with $a, c > 0$, we have

$$\rho(aX + b, cY + d) = \rho(X, Y)$$

Proposition 3.2.11 Let X, Y be random variables.

- $\rho(X, X) = 1$
- $\rho(X, -X) = -1$
- X, Y are uncorrelated if $\rho(X, Y) = 0$

3.3 Moments and Moment Generating Functions

Definition 3.3.1 (Moments)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Define the n th moment of X to be

$$\mu_n = E[X^n]$$

Definition 3.3.2 (Moment Generating Function)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Define the moment

generating function of X to be the function $M_X : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$M_X(t) = E[e^{tX}]$$

whenever the expectation is defined.

Proposition 3.3.3 Assume that M_X exists in a neighbourhood of 0, that is, there exists $\epsilon > 0$ such that for all $t \in (-\epsilon, \epsilon)$ we have $M_X(t) < \infty$. Then for all $k \in \mathbb{N}$ the k th moment of X exists, and

$$E[X^k] = \frac{d^k}{dt^k} M_X(t) \Big|_{t=0}$$

Proof We have that $E[X^k] = \int_{-\infty}^{\infty} x^k f_X(x) dx$ for any continuous cumulative probability. On the other hand,

$$\begin{aligned} \frac{d^k}{dt^k} M_X(t) \Big|_{t=0} &= \frac{d^k}{dt^k} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \Big|_{t=0} \\ &= \int_{-\infty}^{\infty} \frac{\partial^k}{\partial t^k} e^{tx} f_X(x) dx \Big|_{t=0} \\ &= \int_{-\infty}^{\infty} x^k e^{tx} f_X(x) dx \Big|_{t=0} \\ &= \int_{-\infty}^{\infty} x^k f_X(x) dx \end{aligned}$$

■

Proposition 3.3.4

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then the following are true regarding the moment generating function.

- For any $a, b \in \mathbb{R}$, we have $M_{aX+b}(t) = e^{tb} M_X(at)$
- If X, Y are independent, then $M_{X+Y} = M_X \cdot M_Y$

Proposition 3.3.5 Let X, Y be two random variables. Assume that the moment generating functions of X, Y exists and are finite on an interval of the form $(-\epsilon, \epsilon)$. Assume further that $M_X(t) = M_Y(t)$ for all $t \in (-\epsilon, \epsilon)$. Then X, Y have the same distribution.

3.4 Conditional Expectations

Definition 3.4.1 (Conditional Expectations on Subalgebras) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Let $\mathcal{H}$ be a σ -subalgebra of $\mathcal{F}$. Define $E[X | \mathcal{H}] : \Omega \rightarrow \mathbb{R}$ to be a random variable such that the following are true.

- $E[X | \mathcal{H}]$ is $\mathcal{H}$ -measurable.
- For any $A \in \mathcal{H}$, we have $E[X \cdot 1_A] = E[E[X | \mathcal{H}] \cdot 1_A]$

Lemma 3.4.2 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Let $\mathcal{H}$ be a σ -subalgebra of $\mathcal{F}$. Then the random variable $E[X | \mathcal{H}]$ exists and is unique up to almost surely equality.

Lemma 3.4.3 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Let $\mathcal{H}$ be a σ -subalgebra of $\mathcal{F}$. Then the following are true.

- Stability: If X is $\mathcal{H}$ -measurable, then $E[XY | \mathcal{H}] = X E[Y | \mathcal{H}]$.
- Independence: If $\sigma(X)$ and $\mathcal{H}$ are independent, then $E[X | \mathcal{H}] = E[X]$.

Definition 3.4.4 (Conditional Expectation on Random Variables) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Define the conditional expectation of X on Y to be

$$E[X | Y] = E[X | \sigma(Y)]$$

Definition 3.4.5 (Conditional Density) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Define the conditional density of X on the event $\{\omega \in \Omega | Y(\omega) = y\}$ by

$$f_{X | Y}(x, y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Lemma 3.4.6 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Then we have

$$E[X | Y](\omega) = E[X | Y = Y(\omega)] = \int_{-\infty}^{\infty} x f_{X | Y}(x, Y(\omega)) dx$$

4 Convergence of Random Variables

4.1 Different Notions of Convergences

Definition 4.1.1 (Convergence in Mean Square) We say that a sequence of random variables $X_1, X_2, \dots$ converges in mean square to a random variable X if

$$\lim_{n \rightarrow \infty} E[(X_n - X)^2] = 0$$

Definition 4.1.2 (Convergence in Probability) We say that a sequence of random variables $X_1, X_2, \dots$ converges in probability to a random variable X if for every $\epsilon > 0$, we have

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$$

Definition 4.1.3 (Convergence in Distribution) We say that a sequence of random variables $X_1, X_2, \dots$ converges in distribution to a random variable X if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for every x in the set $C = \{x \in \mathbb{R} : F_X \text{ is continuous at } x\}$.

Proposition 4.1.4 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \rightarrow \mathbb{R}$ be a sequence of random variables for $n \in \mathbb{N} \setminus \{0\}$. Let $X : \Omega \rightarrow \mathbb{R}$ also be a random variable. Then the following are true.

- If X_n converges in mean square to X , then X_n converges in probability to X .
- If X_n converges in probability to X , then X_n converges in distribution to X .

4.2 Law of Large Numbers

Theorem 4.2.1 (Markov Inequality) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. If $E[X] < \infty$, then we have

$$P(|X| \geq a) \leq \frac{E[X]}{a}$$

for any $a > 0$.

Theorem 4.2.2 (Chebyshev Inequality) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. If $E[X^2] < \infty$, then we have

$$P(|X| \geq a) \leq \frac{E[X^2]}{a^2}$$

for any $a > 0$.

Theorem 4.2.3 (Weak law of large numbers) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \rightarrow \mathbb{R}$ for $n \in \mathbb{N} \setminus \{0\}$ be a sequence of independently identically distributed random variables with mean μ . Let $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then we have

$$\lim_{n \rightarrow \infty} P(|S_n - \mu| > \epsilon) = 0$$

for all $\epsilon > 0$. In other words, $(S_n)_{n \in \mathbb{N} \setminus \{0\}}$ converges in probability to μ .

Theorem 4.2.4 (Law of large numbers in mean square) Let $X_1, X_2, \dots$ be a sequence of inde-

pendent random variable, each with mean μ and variance σ^2 . Then

$$\lim_{n \rightarrow \infty} \frac{X_1 + \cdots + X_n}{n} \rightarrow \mu$$

in mean square.

4.3 Central Limit Theorem

Theorem 4.3.1 (Central Limit Theorem) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \rightarrow \mathbb{R}$ be a sequence of independent and identically distributed random variables for $n \in \mathbb{N} \setminus \{0\}$, each with mean μ and variance $\sigma^2 \neq 0$. Let $S_n = X_1 + \cdots + X_n$. Then the standardized version of S_n ,

$$Z_n = \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution as $n \rightarrow \infty$ to a Gaussian random variable with mean 0 and variance 1. That is,

$$\lim_{n \rightarrow \infty} P(Z_n \leq x) = \lim_{n \rightarrow \infty} F_{Z_n}(x) = F_Y(y) = \int_{-\infty}^x -\frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

5

5.1

Definition 5.1.1 (Random Samples)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be random variables. We say that they are a random sample of size n from the population f if the following are true.

- $X_1, \dots, X_n$ are pairwise independent.
- The probability density (mass) function is given by $f_{X_i} = f_X$.

The fact that their probability density functions are equal imply that the random variables are identically distributed.

Definition 5.1.2 (Statistics)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. A statistic of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition 5.1.3 (Sample Mean)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample mean of the random sample to be the statistic defined by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Definition 5.1.4 (Sample Variance)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample variance of the random sample to be the statistic defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Definition 5.1.5 (Sample Standard Deviation)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample standard deviation of the random sample to be the statistic defined by

$$S = \sqrt{S^2}$$

6 Principles of Data Reduction

6.1 Sufficiency Principle

Suppose that $(\Omega, \mathcal{F}, P)$ is a probability space. Often we do not know what P is actually like. To probe the properties of P , we may use statistics $T(X_1, \dots, X_n)$ on the probability space.

A sufficient statistic T is a statistic where, informally speaking, all information about P must come from T .

Definition 6.1.1 (Sufficient Statistics)

Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. We say that T is sufficient if

$$P_{\theta_1}(X^{-1}(A) \mid T^{-1}(t)) = P_{\theta_2}(X^{-1}(A) \mid T^{-1}(t))$$

for all $\theta_1, \theta_2 \in I$ and all $A \in \mathcal{F}$ and all $t \in \mathbb{R}$.

Lemma 6.1.2

Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. If the ratio

$$\frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{T(X_1, \dots, X_n)}(x_1, \dots, x_n)}$$

does not depend on θ , then T is a sufficient statistics.

Proof

Firstly notice that if $t \notin \text{im}(T)$ then the probability is 0. Hence we only need to consider the case that $T(X_1, \dots, X_n) = T(x_1, \dots, x_n)$. Denote $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{x} = (x_1, \dots, x_n)$. We have that

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{P_\theta((\mathbf{X} = \mathbf{x}) \cap (T(\mathbf{X}) = T(\mathbf{x})))}{P_\theta(T(\mathbf{X}) = T(\mathbf{x}))} \\ &= \frac{P_\theta(\mathbf{X} = \mathbf{x})}{P_\theta(T(\mathbf{X}) = T(\mathbf{x}))} \\ &= \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{T(X_1, \dots, X_n)}(x_1, \dots, x_n)} \end{aligned}$$

so if the ratio does not depend on θ , we are done. ■

Example 6.1.3

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample of Bernoulli distribution with parameter $\theta \in (0, 1)$ (Note that for each θ , $P_{X_i} = f_{X_i}$ is different since the formula for f_{X_i} depends on θ). Then $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ .

Proof

Let $\theta \in \mathbb{R}$. We use the above lemma. We have

$$\begin{aligned}
 P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{T(X_1, \dots, X_n)}(x_1, \dots, x_n)} \\
 &= \frac{\prod_{i=1}^n f_{X_i}(x_i)}{f_{X_1 + \dots + X_n}(x_1, \dots, x_n)} \\
 &= \frac{\prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\
 &= \frac{\theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\
 &= \frac{1}{\binom{n}{\sum_{i=1}^n x_i}}
 \end{aligned}$$



Theorem 6.1.4 (Neyman-Fisher Factorization Theorem)

Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. Let f_θ be the probability density function of P_θ . Then T is sufficient if and only if there exists probability density functions g_θ for each θ such that

$$f_\theta(x) = g_\theta(T(x))h(x)$$

for all $x \in \Omega$ and all $\theta \in I$.

Definition 6.1.5 (Minimal Sufficient Statistics)

Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let T be a sufficient statistic of the random sample. We say that T is minimal if for any other sufficient statistic T' , we have T is a function of T' .

Definition 6.1.6 (Ancillary Statistics)

Definition 6.1.7 (Complete Statistics)

Theorem 6.1.8 (Basu's Theorem)

6.2 Likelihood Principle

6.3 Conditionality Principle

6.4 Equivariance Principle

7 Estimators

Suppose some real life problem is modelling statistically by the probability space $(\Omega, \mathcal{F}, P)$. To probe what P looks like, we sample the space using random variables $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$. Based on the sample, we may infer that P may have parameter $\hat{\theta}$ (naturally $\hat{\theta}$ should be a function of $X_1, \dots, X_n$). We may then need to evaluate how accurate $\hat{\theta}$ is compared to the actual parameter θ .

There are two main branches in estimator theory: We need different methods for finding estimators, and we need to evaluate how accurate the estimators are.

There are also two types of estimators: Point estimators and interval estimators.

7.1 Finding Point Estimators

Definition 7.1.1 (Point Estimators)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n$ be a random sample. A point estimator of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \rightarrow \mathbb{R}$.

Note: This the same definition of a statistic.

Definition 7.1.2 (Method of Moments)

Definition 7.1.3 (Maximum Likelihood Estimators)

7.2 Evaluating Point Estimators

Definition 7.2.1 (Bias of a Point Estimator)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let $\hat{\theta}$ be a point estimator of the random sample. Define the bias of $\hat{\theta}$ to be

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Definition 7.2.2 (Unbiased Point Estimators)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let $\hat{\theta}$ be a point estimator of the random sample. We say that $\hat{\theta}$ is unbiased if $\text{Bias}(\hat{\theta}) = 0$.

Definition 7.2.3 (Mean Squared Error)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let $\hat{\theta}$ be a point estimator of the random sample. Define the mean squared error of $\hat{\theta}$ to be

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Theorem 7.2.4 (The Bias Variance Trade Off)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let $\hat{\theta}$ be a point estimator of the random sample. Then we have

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$$

8 Hypothesis Testing